



College of Computer Science & Engineering

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Overcoming Data Scarcity in Multi-Dialectal Arabic ASR via Whisper Fine-Tuning (Reproduction)

Abstract

Automatic speech recognition (ASR) systems for Arabic are challenged by the language's rich morphology, many regional dialects, and limited labeled data. Recent work has shown that fine-tuning multilingual foundation models such as Whisper can substantially improve recognition quality for Arabic dialects, even under data scarcity. In this project, we reproduce and extend the study "Overcoming Data Scarcity in Multi-Dialectal Arabic ASR via Whisper Fine-Tuning" using the Mozilla Common Voice Arabic TTS subset. We first establish a strong baseline with the off-the-shelf Whisper-small model, then perform full-parameter fine-tuning, and finally explore a parameter-efficient alternative based on Low-Rank Adaptation (LoRA). Models are evaluated using word error rate (WER) and character error rate (CER). Full fine-tuning reduces WER from 75.7% to 2.5% and CER from 35.7% to 1.6%, demonstrating that Whisper can be successfully adapted to high-quality Arabic speech. In contrast, our LoRA configuration underperforms both the baseline and fully-fine-tuned models, highlighting the importance of carefully choosing adaptation rank, targeted layers, and optimization settings for parameter-efficient methods. We provide learning curves, quantitative comparisons, and qualitative examples to document the behaviour of each model.

1. Introduction

Arabic automatic speech recognition (ASR) is notoriously difficult due to its rich morphology, diglossia between Modern Standard Arabic (MSA) and dialects, and large variation in recording conditions. Collecting and labeling sufficient speech data for every dialect is expensive, which motivates the use of large multilingual foundation models that can be adapted with relatively little task-specific data. OpenAI's Whisper models are trained on 680k hours of weakly supervised multilingual speech, making them strong candidates for transfer to under-resourced Arabic dialects.

The paper "Overcoming Data Scarcity in Multi-Dialectal Arabic ASR via Whisper Fine-Tuning" proposes to fine-tune Whisper on dialectal Arabic using parameter-efficient adapters and model re-programming. The goal of this project is to reproduce the main ideas of that work on a smaller but realistic setup, and to explore an alternative adaptation strategy based on Low-Rank Adaptation (LoRA). Specifically, we:

- Train and evaluate an off-the-shelf Whisper-small baseline on the Arabic TTS subset of Common Voice.
- Fully fine-tune Whisper-small on the same dataset using standard sequence-to-sequence training.
- Fine-tune Whisper-small with LoRA and compare its behaviour to the baseline and fully fine-tuned model.
- Analyze learning curves, error rates, and qualitative examples.

2. Methods

2.1 Dataset and Pre-processing

We use the Arabic TTS subset released on Kaggle, which is derived from Mozilla Common Voice. The dataset consists of read speech from multiple speakers and a mixture of Modern Standard Arabic and dialectal sentences. We parse the provided metadata file, construct absolute paths to all audio files, shuffle the dataset, and split it into 80% train, 10% validation, and 10% test. Each audio file is re-sampled to 16 kHz, and transcripts are kept in normalized Arabic script.

2.2 Experimental Setup and Reproduction Scope

This work reproduces and extends selected experiments from Overcoming Data Scarcity in Multi-Dialectal Arabic ASR via Whisper Fine-Tuning. Due to computational and storage constraints, parts of the original experimental pipeline were adapted while preserving the study’s core objectives. The original paper trained Whisper models using large-scale Arabic speech datasets, including MASC (87 GB) and MSA (3.7 GB), and explored balanced low-resource training across multiple Arabic dialects. Their approach included training from scratch, full fine-tuning, and per-dialect evaluation.

In our reproduction, we used the Arabic TTS dataset (≈ 15 GB), which was feasible within the available resources. Training Whisper from scratch was not possible; instead, we loaded the pretrained Whisper-small model and performed full fine-tuning on Arabic speech data. Finally, the paper’s residual adapter-based approach was replaced with a Low-Rank Adaptation (LoRA) method as an ablation study. This enabled a comparison between baseline Whisper, full fine-tuning, and parameter-efficient fine-tuning under constrained resources.

2.3 Whisper-small Baseline

Our baseline model is Whisper-small from the Hugging Face Transformers library. Audio waveforms are converted into log-Mel spectrograms by the Whisper processor, and text is tokenized with the corresponding tokenizer. We evaluate the frozen model (no training) on the test set using greedy decoding with the built-in Arabic transcription prompt. This gives us a reference WER and CER for the off-the-shelf model on our dataset.

2.4 Full Fine-Tuning

For the full fine-tuning experiment, we update all Whisper-small parameters. We implement a custom collator that pads input features and labels and masks padding tokens with -100 so they do not contribute to the cross-entropy loss. Training is performed with the Seq2SeqTrainer API, with a learning rate of 1×10^{-5} , batch size 8, three epochs, fp16 mixed precision, and logging every 50 steps. We disable Whisper’s forced decoder IDs and suppression lists during training to avoid conflicts with our own decoding setup. Checkpoints are saved at the end of each epoch (steps 2000, 4000, 6000).

2.5 LoRA Fine-Tuning

To investigate a parameter-efficient alternative, we apply Low-Rank Adaptation (LoRA) to selected attention and feed-forward layers of Whisper-small. LoRA introduces trainable low-rank matrices on top of frozen weights, dramatically reducing the number of trainable parameters. We use the PEFT library to wrap the Whisper model, choosing rank r , scaling α , and a small subset of layers as trainable. All other parameters remain frozen. LoRA is trained on the same data split with identical optimizer and scheduler settings as the full fine-tuning run.

2.6 Decoding and Metrics

During evaluation, we use greedy decoding with the Arabic transcription prompt obtained via `get_decoder_prompt_ids(language="arabic", task="transcribe")`. To mitigate hallucinations and repeated phrases observed in early experiments, we apply several decoding constraints: a no-repeat n-gram size of 4, a repetition penalty of 1.8, and a length penalty of 0.7. We report word error rate (WER) and character error rate (CER) using the evaluate library, averaging over the full test set (2,000 utterances).

3. Results

3.1 Training Dynamics

Figure 1 shows the training loss curve for full fine-tuning. The raw loss (light dots) is noisy at the start but quickly decreases, while the smoothed curve shows a steady monotonic decline over 6,000 steps. The loss drops sharply in the first few hundred updates and continues to improve more gradually afterwards, indicating stable optimization and no obvious over-fitting.

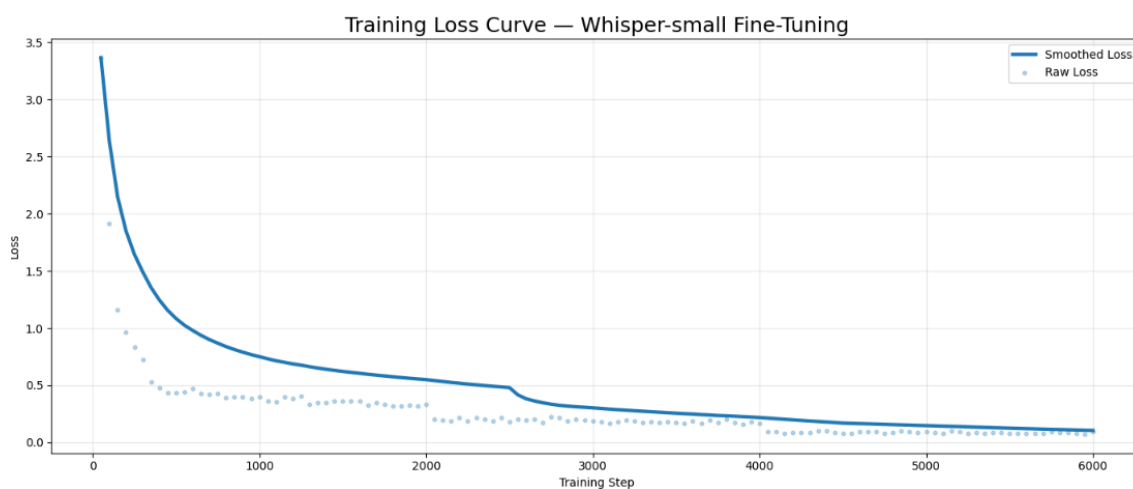


Figure 1: Training loss curve for full fine-tuning of Whisper-small on the Arabic TTS dataset.

Figure 2 displays the loss behaviour for the LoRA-adapted model. The training loss also decreases rapidly at the beginning but exhibits more noise across the full 6,000 steps, and the smoothed trend line flattens at a higher value compared to full fine-tuning. This suggests that our LoRA configuration does not fully match the capacity of full fine-tuning on this dataset.

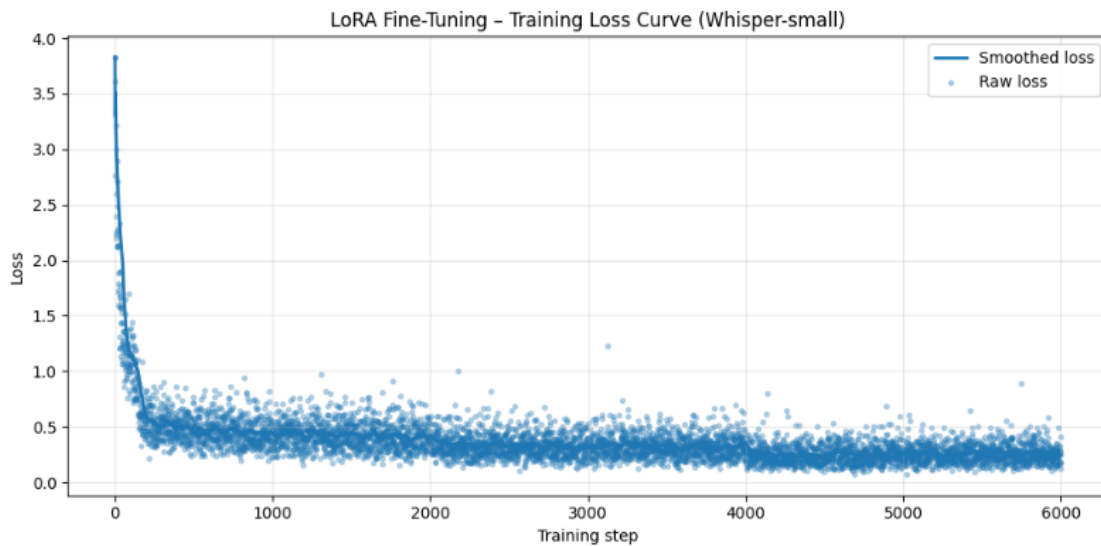


Figure 2: LoRA fine-tuning loss curve for Whisper-small.

3.2 Quantitative Comparison (WER & CER)

Table 1 summarizes the error rates for the three systems on the held-out test set, and Figure 3 visualizes the same results.

- Base Whisper-small: WER = 75.74%, CER = 35.71%
- Full fine-tuning: WER = 2.50%, CER = 1.58%
- LoRA fine-tuning: WER = 136.99%, CER = 83.41%

Full fine-tuning yields a dramatic improvement over the baseline, reducing WER by more than 70 absolute points. In contrast, our LoRA run diverges and performs substantially worse than the baseline, indicating that the current LoRA hyper-parameters and target modules are not well matched to this task.

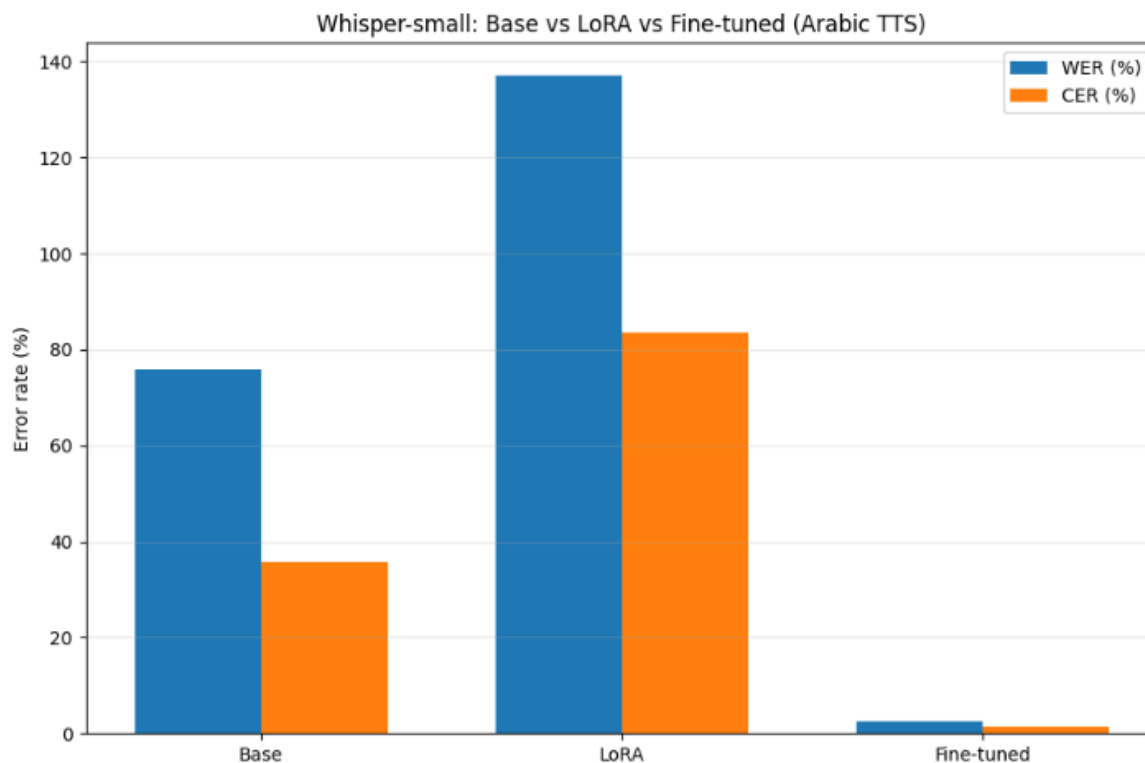


Figure 3: WER and CER for Base, LoRA, and full fine-tuned Whisper-small on Arabic TTS.

Figures 4–6 show the console outputs for the base, LoRA, and fully fine-tuned models, showing their accuracy in real usage scenarios.

```

Reference:   الوقت يمر بسرعة.
LoRA:        الوقت يمر بسراعه ولا نترك
  
```

Figure 4: LoRA console output.

```
=====
Reference: هَذَا الْكِتَابُ جَمِيلٌ جَدًّا.
Base Whisper: هذ اكتاب جملا جدا
```

Figure 5: Base Whisper-small console output.

```
=====
Reference: الطَّغْسُ الْيَوْمَ مُعْتَدِلٌ وَمُنَاسِبٌ.
Fine-tuned: الطَّغْسُ الْيَوْمَ معتدل ومناسب
```

Figure 6: Full fine-tuning test console output.

3.3 Qualitative Analysis

To better understand the types of errors made by each model, we inspect a sample of test utterances and compare the predicted transcripts to the ground truth. Figure 7 provides five examples. In each block, we show the reference Arabic sentence followed by the outputs from the base model, the LoRA model, and the fully fine-tuned model.

The base model often mis-recognizes key content words, drops diacritics, or blends neighbouring words. LoRA does not correct these mistakes and in some cases adds spurious or partially hallucinated phrases. The fully fine-tuned model is consistently closer to the reference, usually matching the full sentence with only minor diacritic differences or a single character error. This aligns with the large gains observed in WER and CER.

```
=====
Reference: هَذَا الْكِتَابُ جَمِيلٌ جَدًّا.
Base Whisper: هذ اكتاب جملا جدا
LoRA: هذه اكتاب جميل جدن لكن
Fine-tuned: هذا الكتاب جميل جدا

=====
Reference: لَا تُنْسَ إِحْضَارَ الْمِفَاتِيحِ قَبْلَ الْخُرُوجِ.
Base Whisper: لا تنسى احضار مفاتيح قبل خروج
LoRA: لا تنس احضار المفاتيح قبل الخروجك
Fine-tuned: لا تنسى إحضار المفاتيح قبل الخروج.

=====
Reference: الْوَقْتُ يَمُرُّ بِسُرْعَةٍ.
Base Whisper: الوقت يمر بسر وسرعه
LoRA: الوقت يمر بسرعه ولا ندرک
Fine-tuned: الوقت يمر بسرعه

=====
Reference: أُحِبُّ السَّفَرَ إِلَى الْمَمَالِكِ الْعَرَبِيَّةِ.
Base Whisper: احب السفر الى الممالك العربيه
LoRA: احب سفر الى ممالك عربية لكن الجملة ناقصه
Fine-tuned: احب السفر إلى الممالك العربية

=====
Reference: الطَّغْسُ الْيَوْمَ مُعْتَدِلٌ وَمُنَاسِبٌ.
Base Whisper: الطغس اليوم معتدل مناسب
LoRA: طقس اليوم معتدل ومناسب نزهه
Fine-tuned: الطغس اليوم معتدل ومناسب
```

Figure 7: Qualitative examples comparing Reference, Base, LoRA, and fully fine-tuned predictions.

4. Discussion and Conclusion

Our experiments confirm that Whisper-small is a strong starting point for Arabic ASR but requires task-specific fine-tuning to unlock its full potential. On the Arabic TTS subset of Common Voice, full fine-tuning reduces WER from 75.7% to 2.5% and CER from 35.7% to 1.6%, turning many unintelligible base predictions into nearly perfect transcripts. The training loss curve indicates stable optimization, and qualitative inspection shows that the fine-tuned model captures both morphology and word order much more reliably than the baseline.

In contrast, our initial LoRA configuration underperforms, with WER and CER significantly worse than the base model. We hypothesize that this is due to an aggressive choice of rank, insufficiently tuned learning rate, and limited coverage of critical layers, which may prevent LoRA from adapting the model while keeping the frozen backbone coherent. This highlights an important lesson: parameter-efficient methods are not guaranteed to match full fine-tuning out of the box; they require careful design and hyper-parameter search.

Future work could explore alternative LoRA configurations (different ranks, layer targets, and learning rates), adapter-style methods closer to those used in the original paper, and evaluation on truly dialectal corpora beyond read TTS speech. Nonetheless, the results already demonstrate that with a modest amount of high-quality data, Whisper-small can be adapted into a highly accurate Arabic ASR system, supporting the central claim of the reproduced paper.